# Continuous-time Controllers

Control Engineering

# Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods

# Continuous-time Controllers

- Control objectives
- PID controllers
- On-off controllers
- Working point
- Proportional band

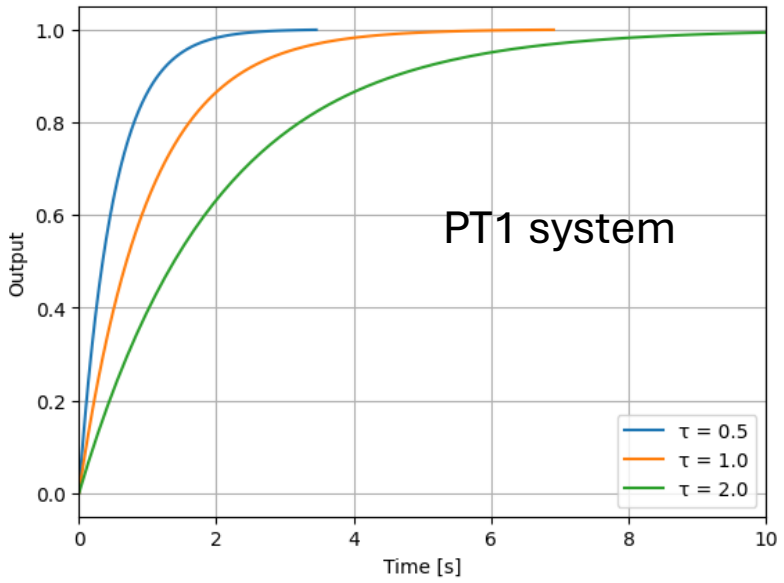
# Control

Models (differential equations) of dynamic systems allow to calculate the effect of inputs on outputs

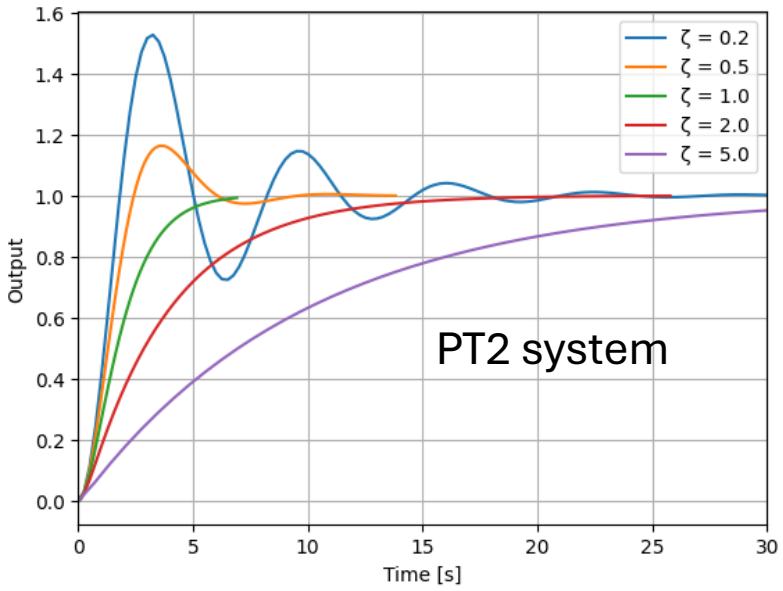
Idea of control: modify the input in order to obtain a desired output

Example: adjust step input to reach a desired steady state ($u_{\text{step}} = \frac{y(\infty)}{K}$)

Problems: response may be too slow, disturbances degrade performance

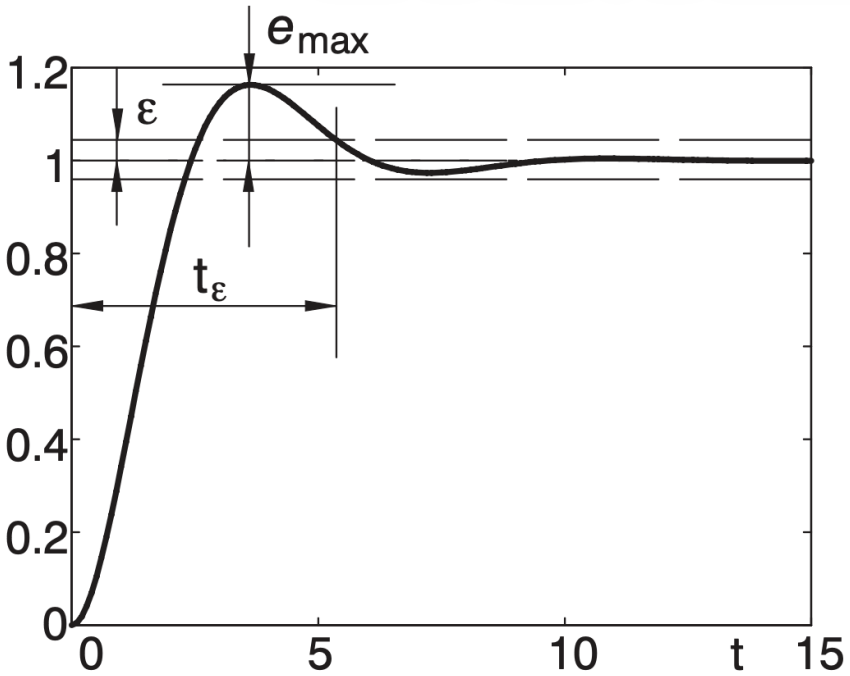


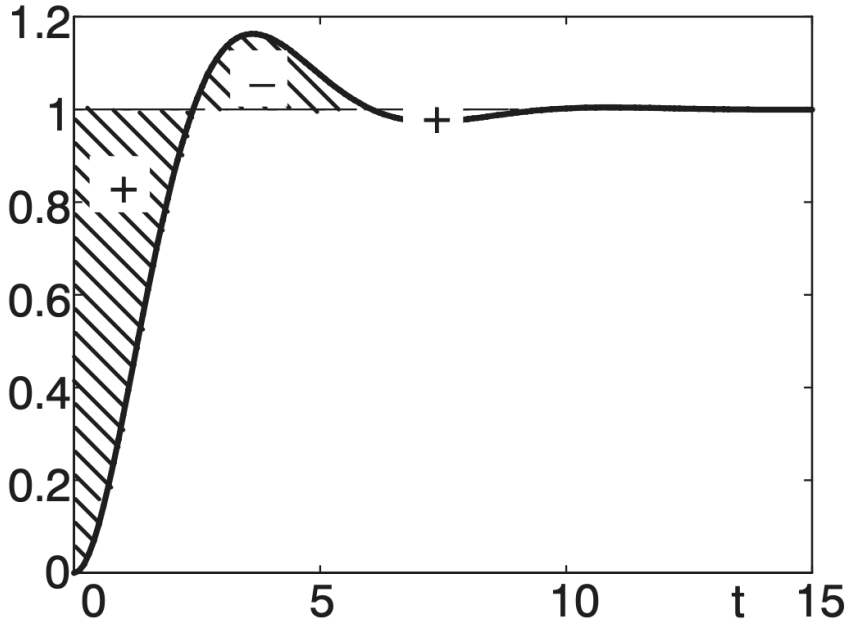

Step responses



# Control Objectives

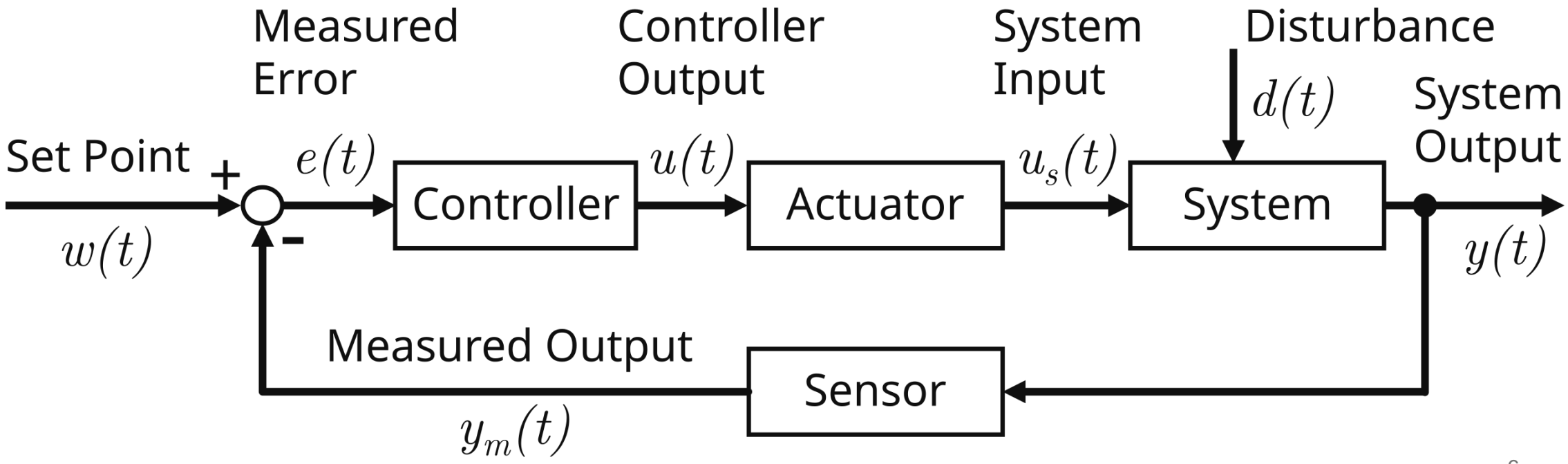
- Stability (bounded output)
- Fast response: rise and settling time
- Low overshoot
- Small steady-state error
- Robustness (insensitivity to model uncertainty and disturbances)



# Control Loop

- Feedback: measure system output, compare it to desired behavior (e.g., steady state), and use the resulting error as input to a controller
- Goal: reduce error



# What Is a Controller?

System itself (subsystem of total control system), implemented in hardware or software

- Input: measured error
- Output: process input (low-energy control signal converted into physical action by actuator)

Hardware controllers

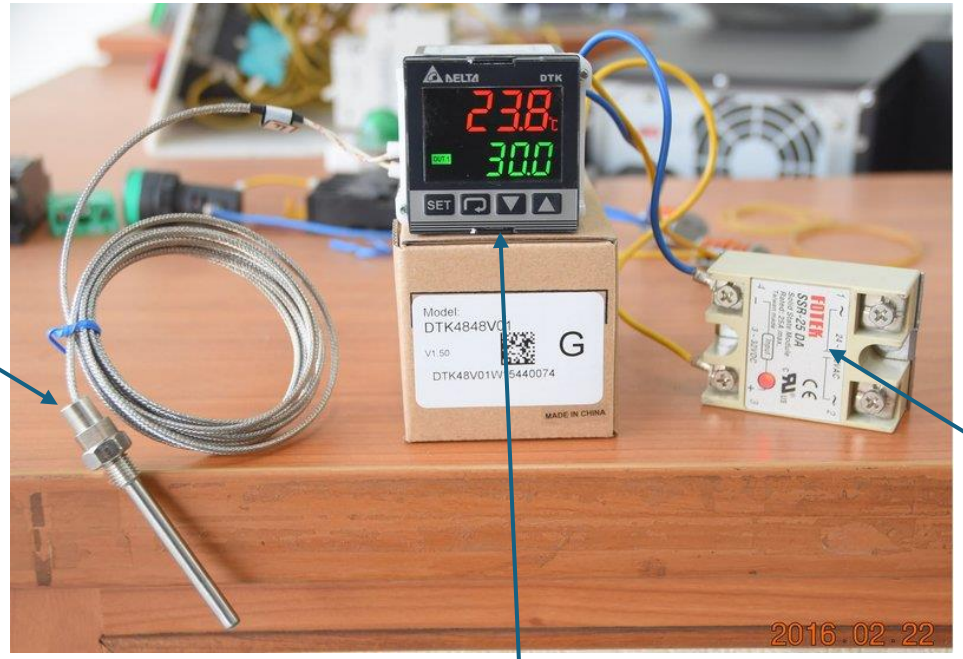
- Fast, deterministic, limited flexibility
- Examples: mechanical, analog electronic controllers using op-amps, FPGA-based (digital)

Software controllers (use of analog actuators via a digital-to-analog converter)

- Flexible, complex algorithms, sampling delays (dead time)
- Examples: Microcontroller-based, PLCs

# Example: Temperature Control

Sensor:
resistance thermometer to measure temperature (system output)



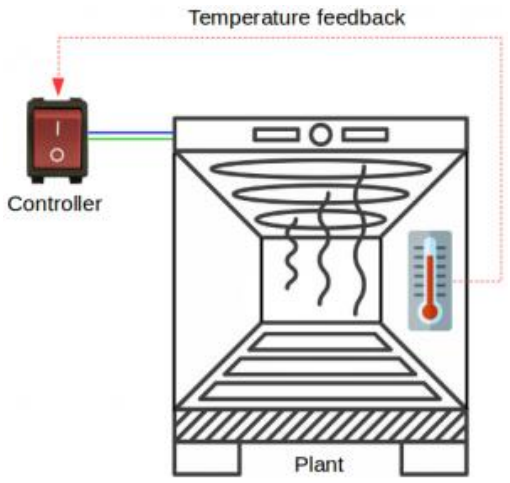

Actuator:
solid-state relay to switch a heater/cooler (plant) on or off

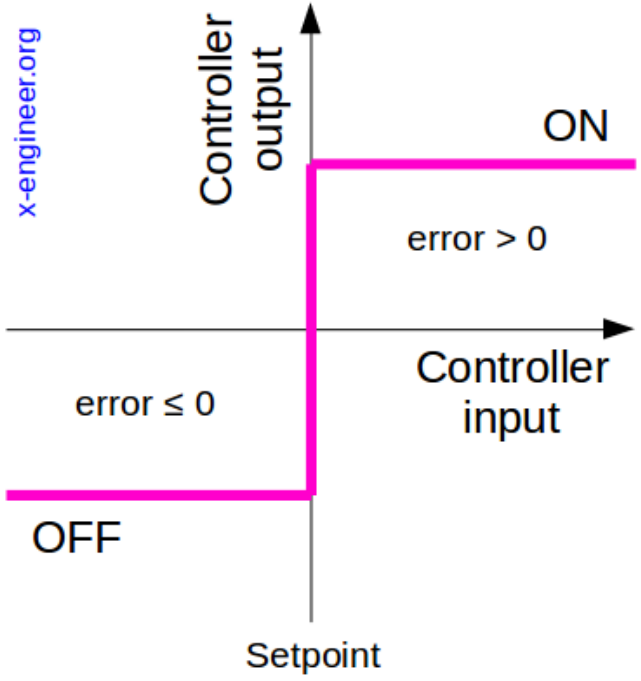
Controller:
digital electronic PID switching controller to activate the relay (controller output)

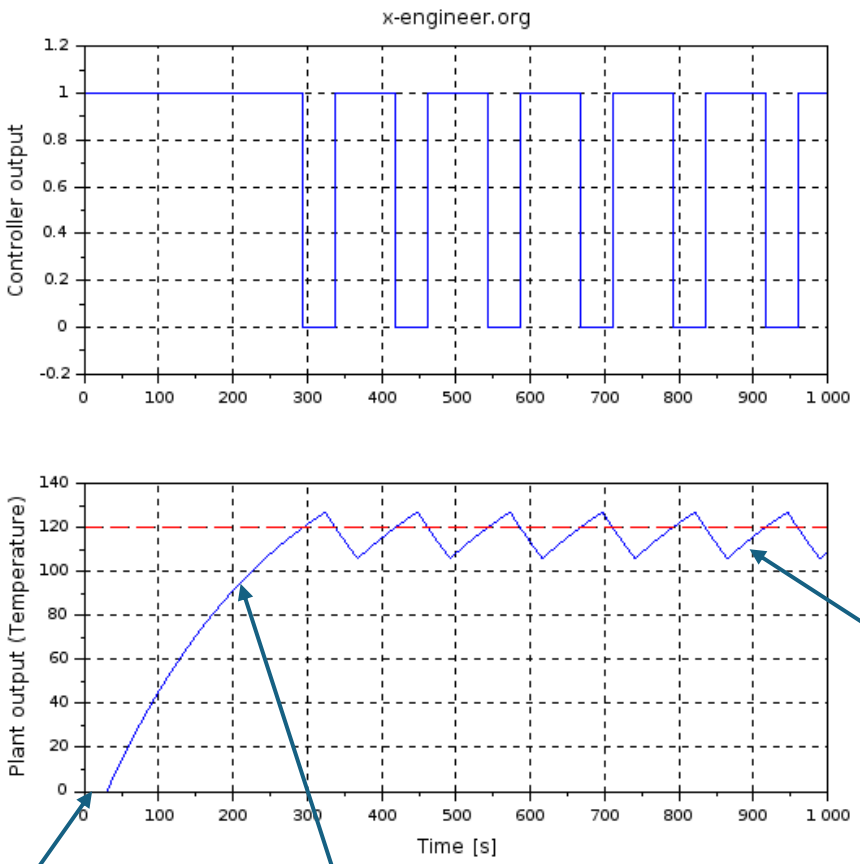
# On-Off Controller

aka bang-bang control

Example: industial oven



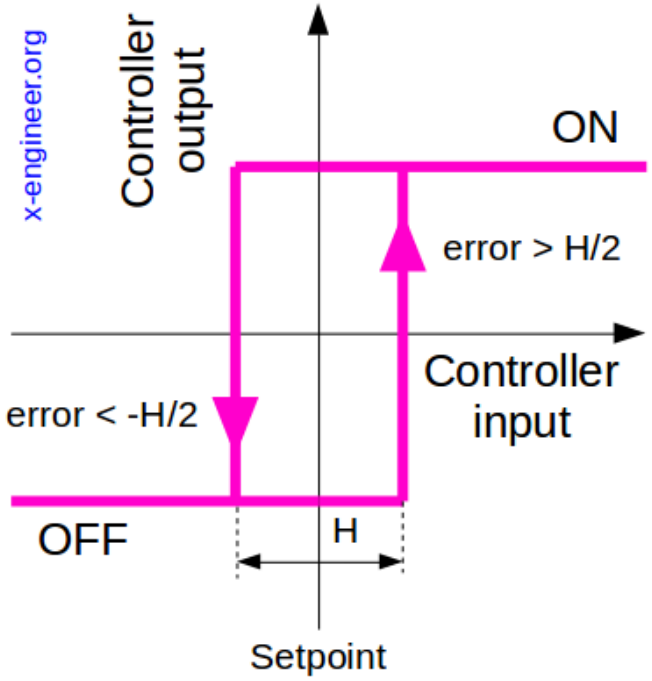

Issue:
High-frequency switching stresses actuator

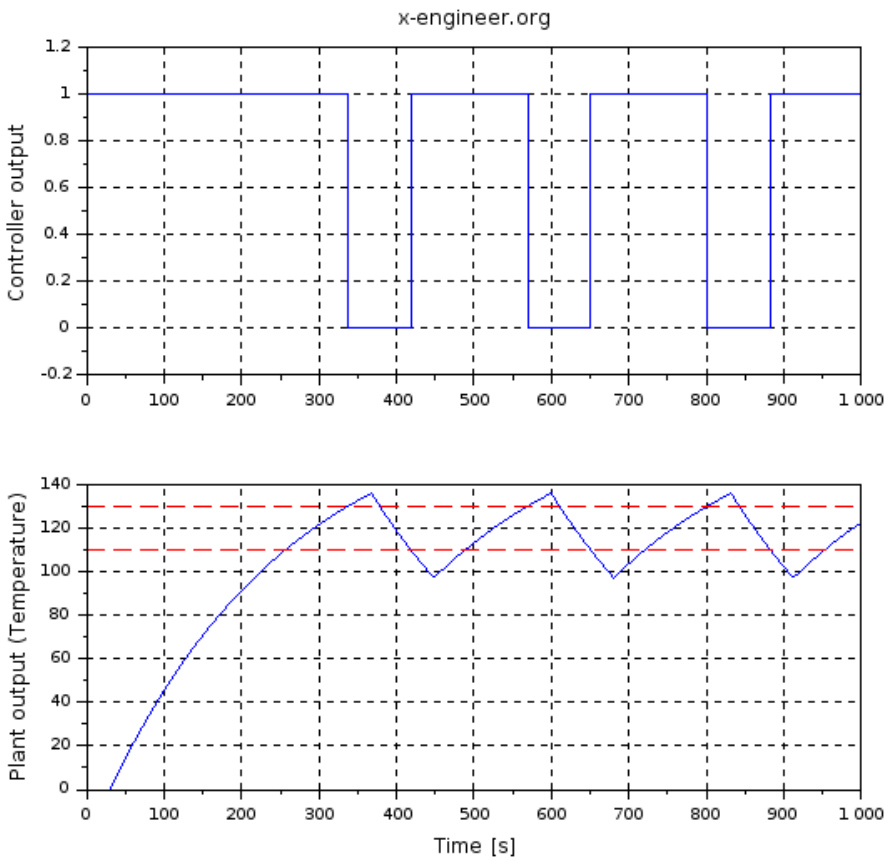
oscillations around setpoint

dead time

first-order process

# On-Off Controller with Hysteresis

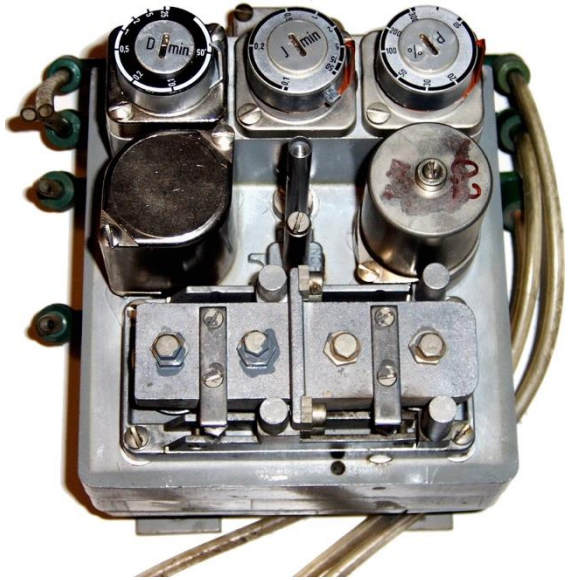



Less switching at the expense of accuracy

On-off controllers are simple and low-cost, but have limited precision.
→ Continous controllers (e.g., PID)

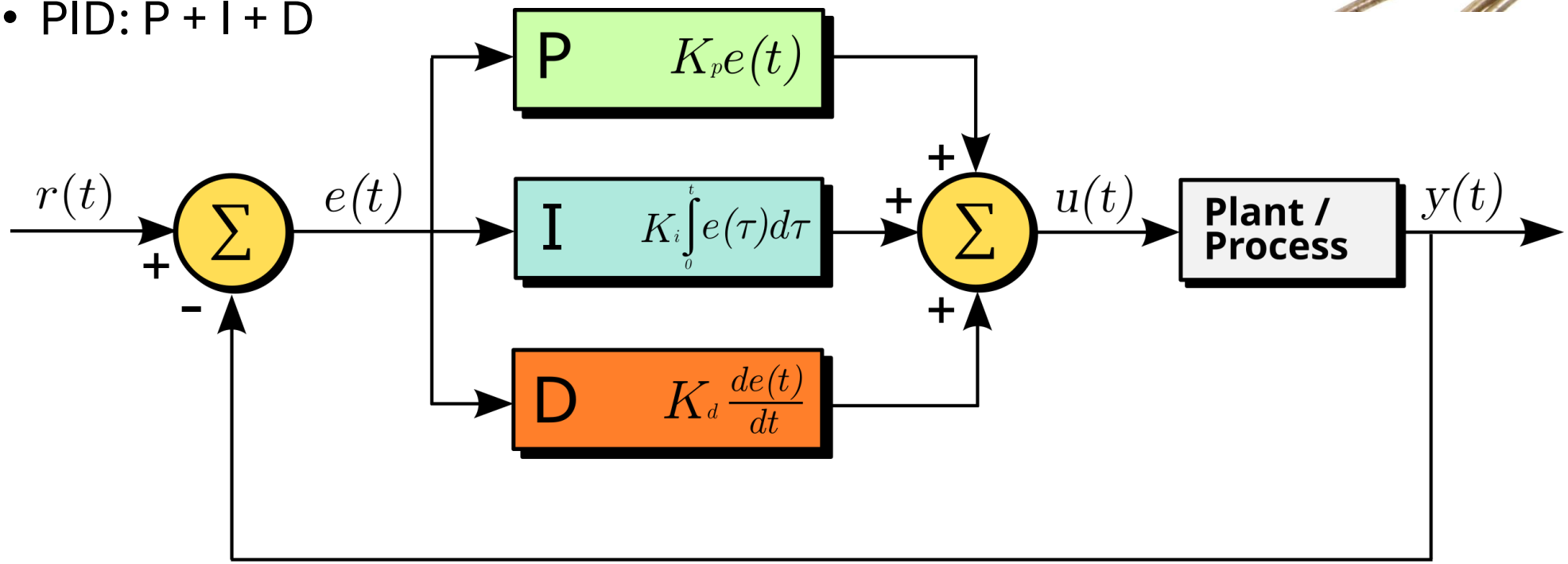
Example: pneumatic PID



# PID Controllers

Most widely used industrial controller

- P
- PI: P + I
- PD: P + D
- PID: P + I + D



# Proportional Control (P)

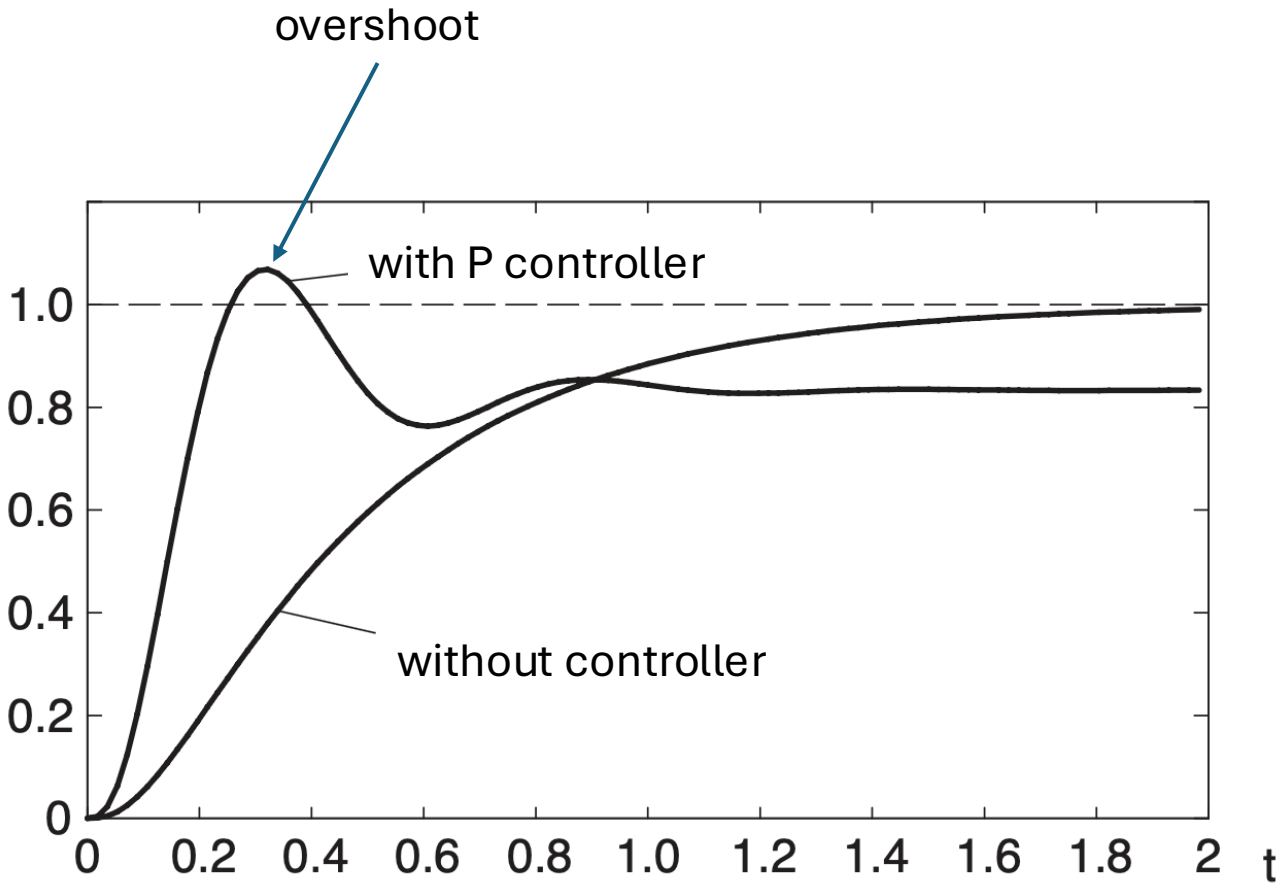
Control input proportional to error (static system):

$$u(t) = K_p \cdot e(t)$$

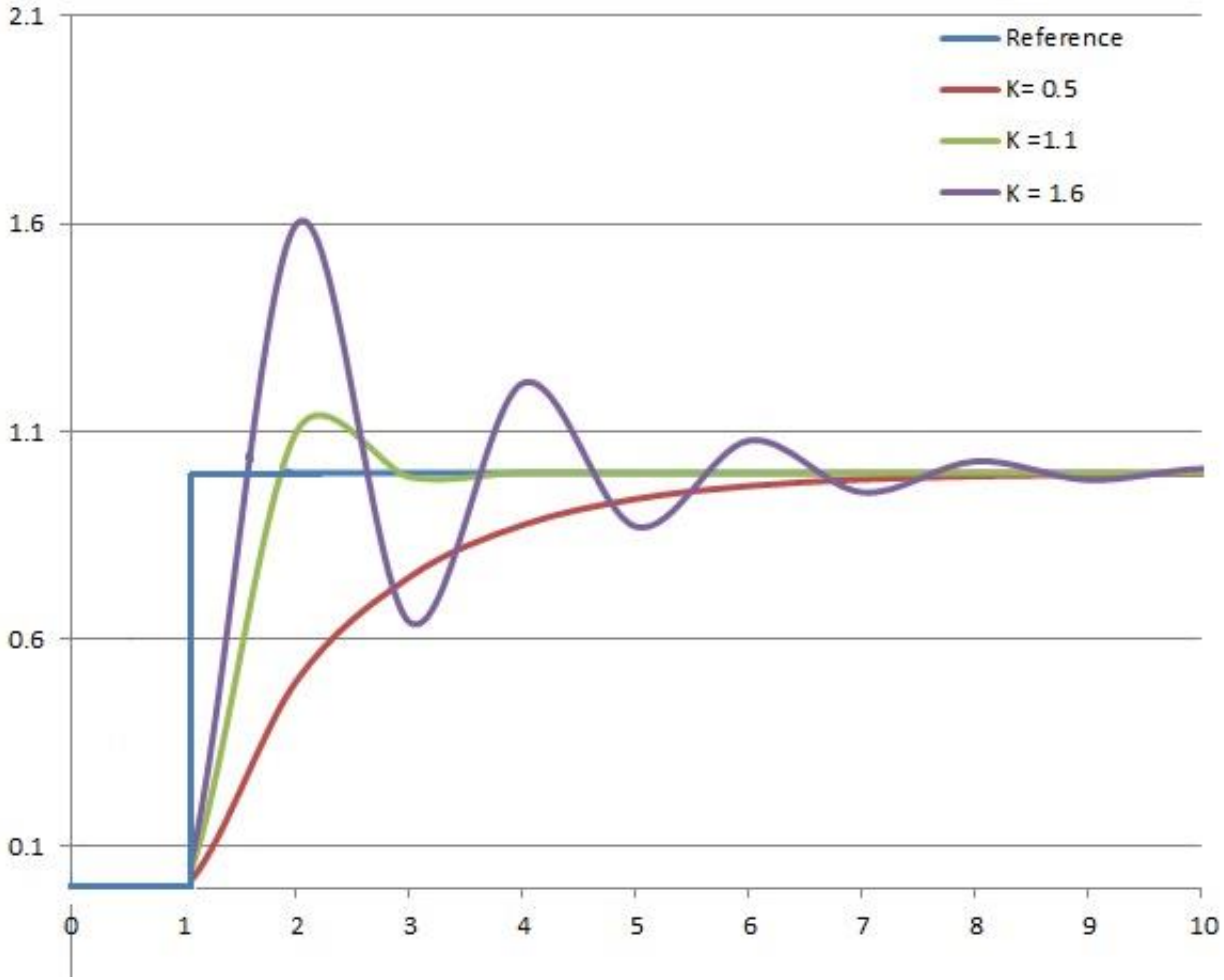
- Increases loop gain → faster response
- Reduces damping → more oscillation
- Steady-state error in many systems:

$$e(t) = 0 \implies u(t) = 0$$

→ $y(t)$ settles at some error, just big enough to produce $u(t)$ balancing the system dynamics (e.g., holding a mass up)



# Effect of Proportional Gain Parameter



# Historical: Centrifugal Governor in a Windmill

Working principle:
wind turns the blades
→ blades turn a shaft (plant)
→ shaft spins a pair of weighted arms (controller)
→ arms mechanically twist the blades (actuator)

Automatic regulation:
faster rotation (controlled variable: angular speed)
→ centrifugal force pushes weights outward (error: deviation from position at reference angular speed)
→ blades are twisted to catch less wind (proportional control) → slower rotation

Steady-state error: need for twisted blades to balance torque from wind and load → speed offset

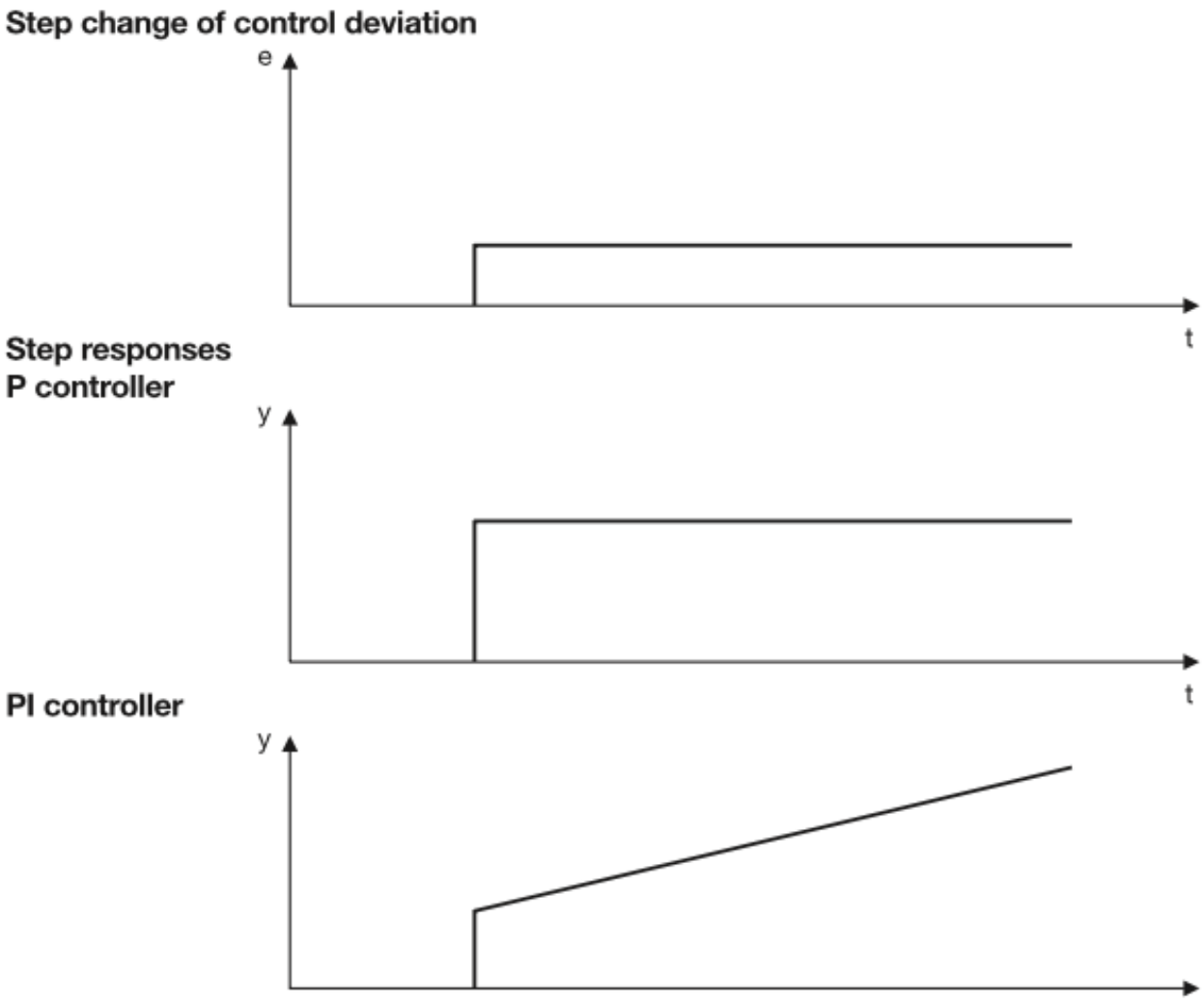
# Integral Control (I)

Integrator accumulating past error (dynamic)

- Eliminates steady-state error to step inputs
- Slower response
- Reduces damping → more oscillation

Order of total system increases (adds memory)

- PT1 → second order
- PT2 → third order

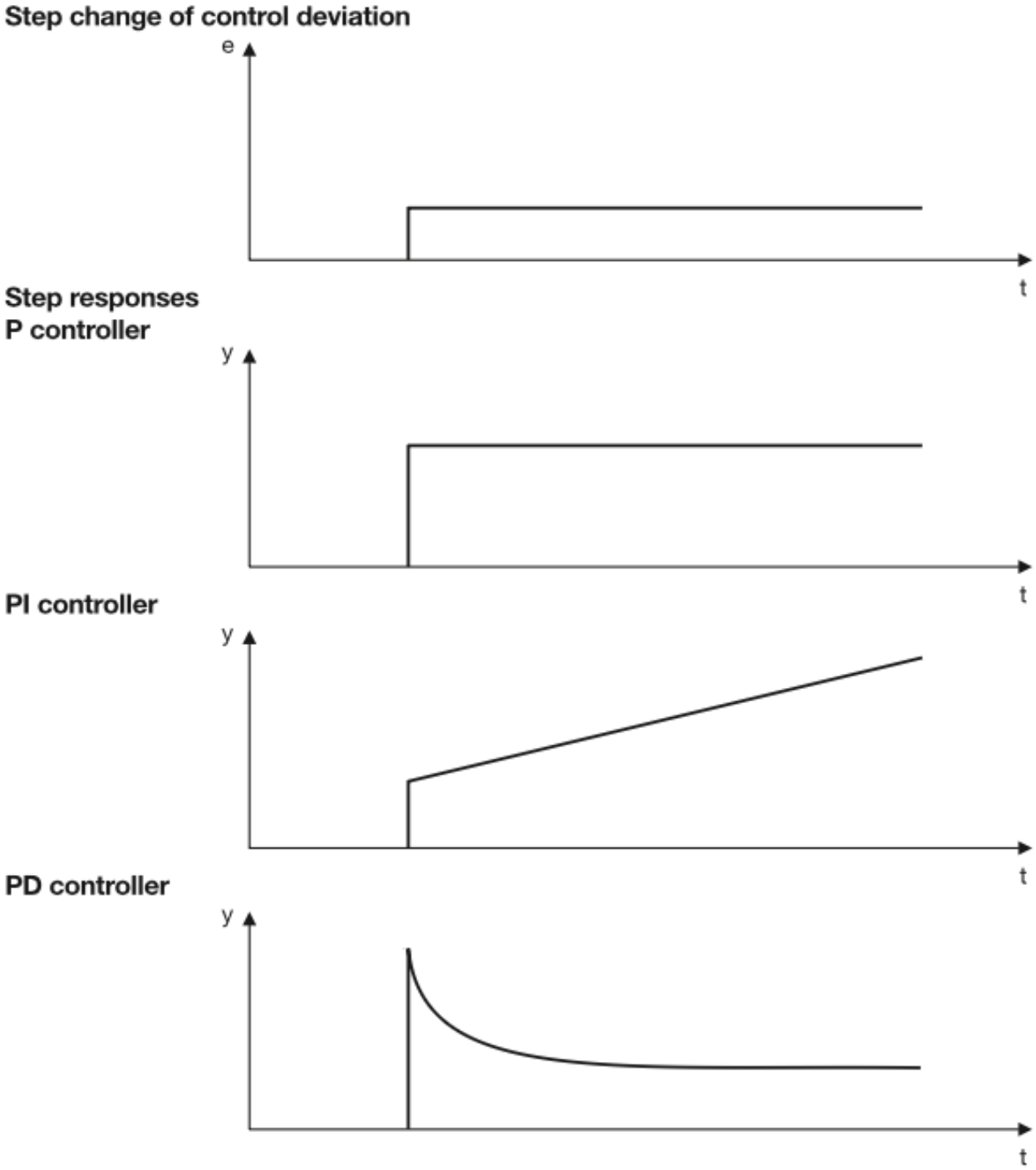


# Derivative Control (D)

Derivative predicting future error based on the rate of change (dynamic)

- Predictive action
- Increases effective damping (keeps system near critical damping): less overshoot, fewer oscillations)
- Derivative action amplifies high-frequency noise → often paired with a low-pass filter

Used less often in practice due to impact on system stability → PI



# PID Controller

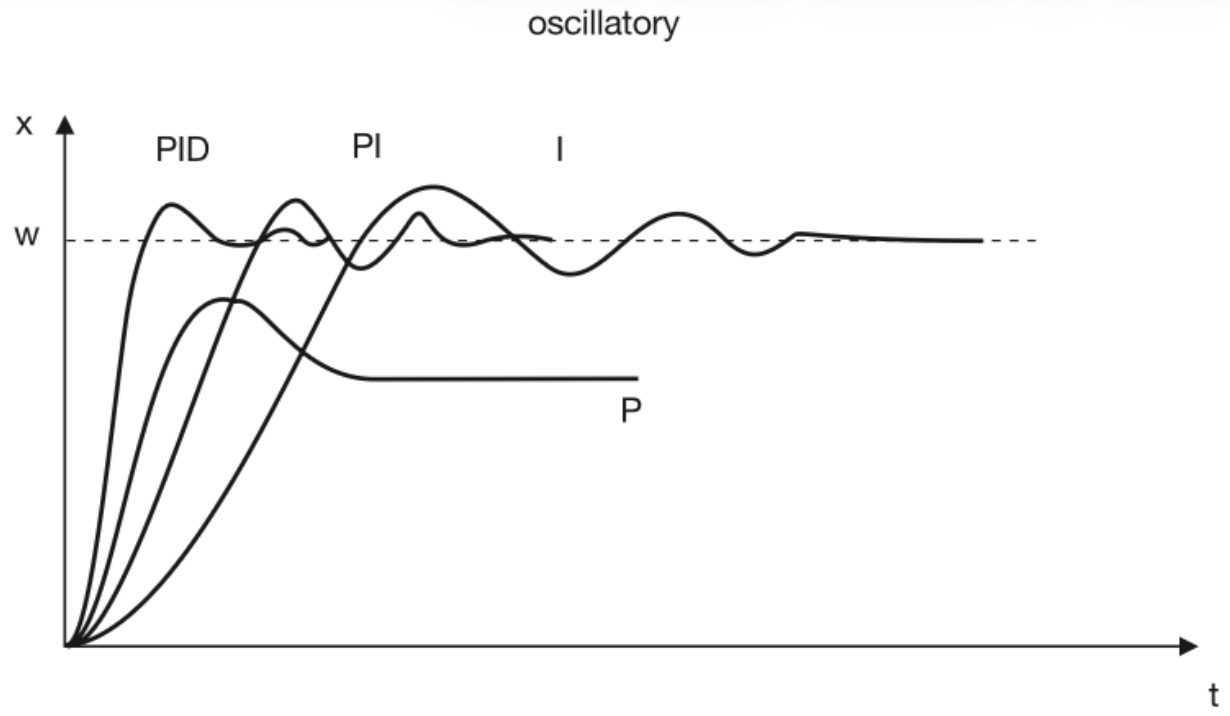
- Simple and intuitive
- Works for many plants
- Easy to tune experimentally
- No model of process needed

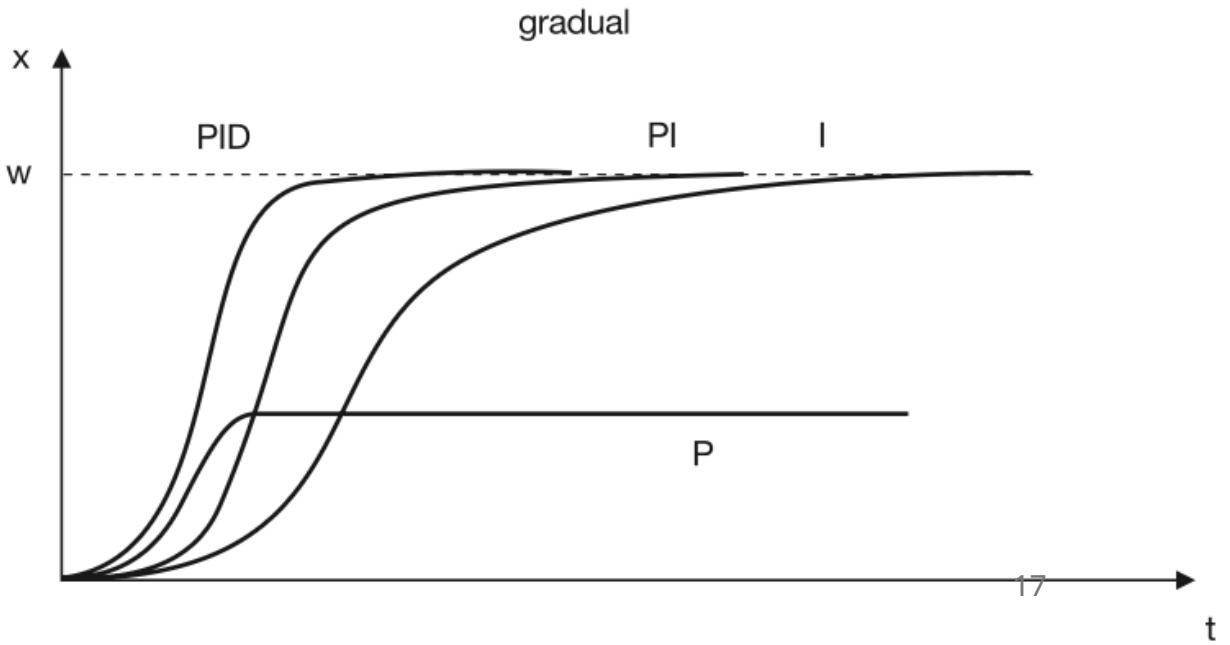
→ ~90% of industrial controllers are PID variants

P = speed

I = accuracy

D = damping



# Time-Domain Performance Trade-Offs

Faster response vs. stability:

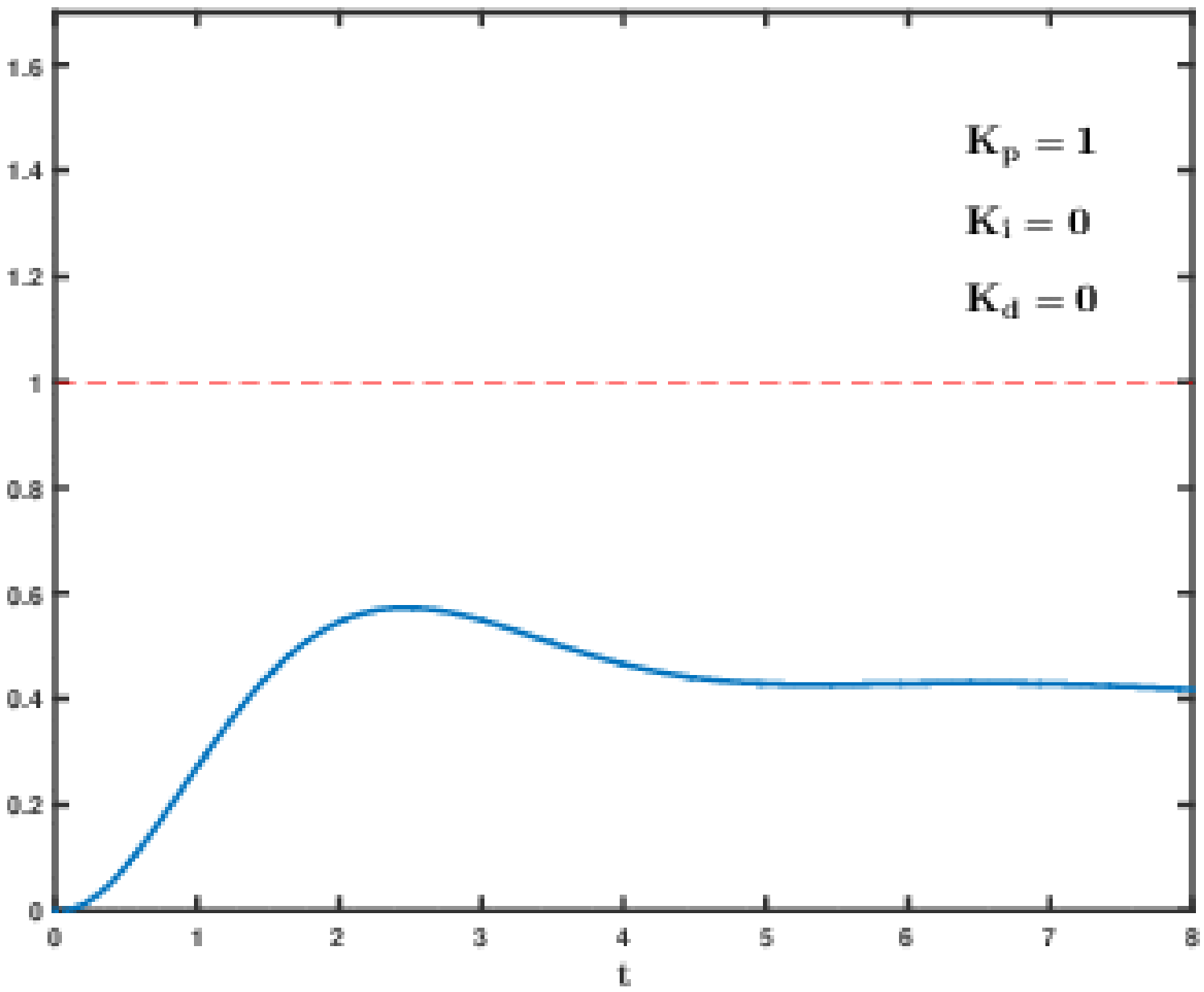
Higher $K_p$ → faster response, but more overshoot and risk of instability

Accuracy vs. robustness:

Higher $K_i$ → higher accuracy, but more sensitive to disturbances

Damping vs. noise sensitivity:

Higher $K_d$ → less overshoot, but amplifies measurement noise



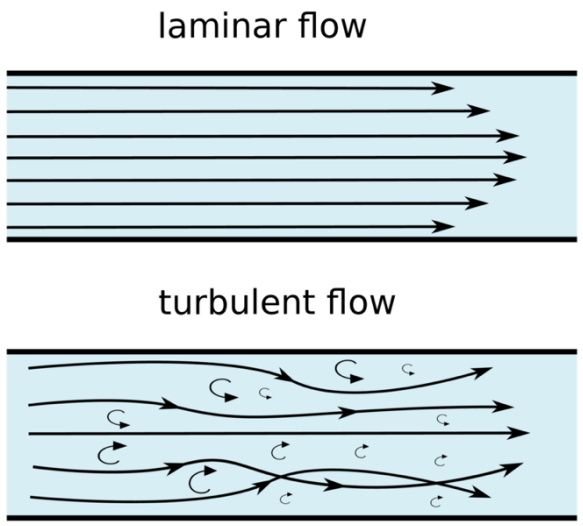

Animated GIF

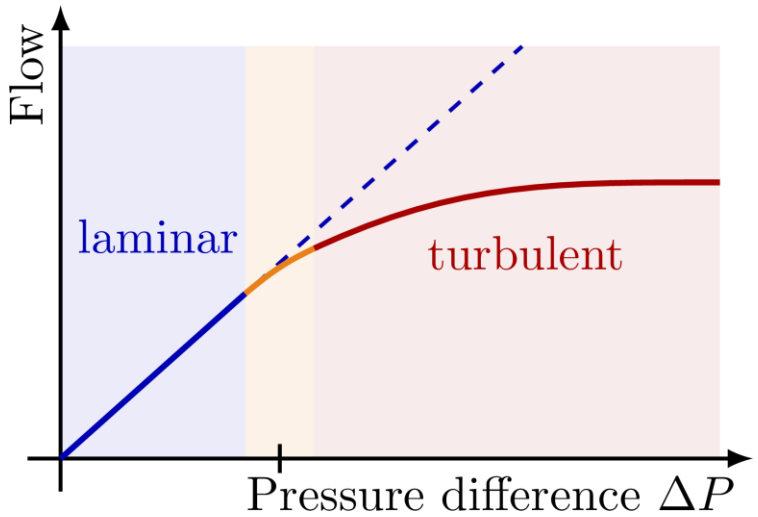
# Nonlinear Systems

Linear models (output proportional to the input) are often used because they are easier to analyze.

However, real systems are typically nonlinear.

Example: flow resistance (laminar vs turbulent)

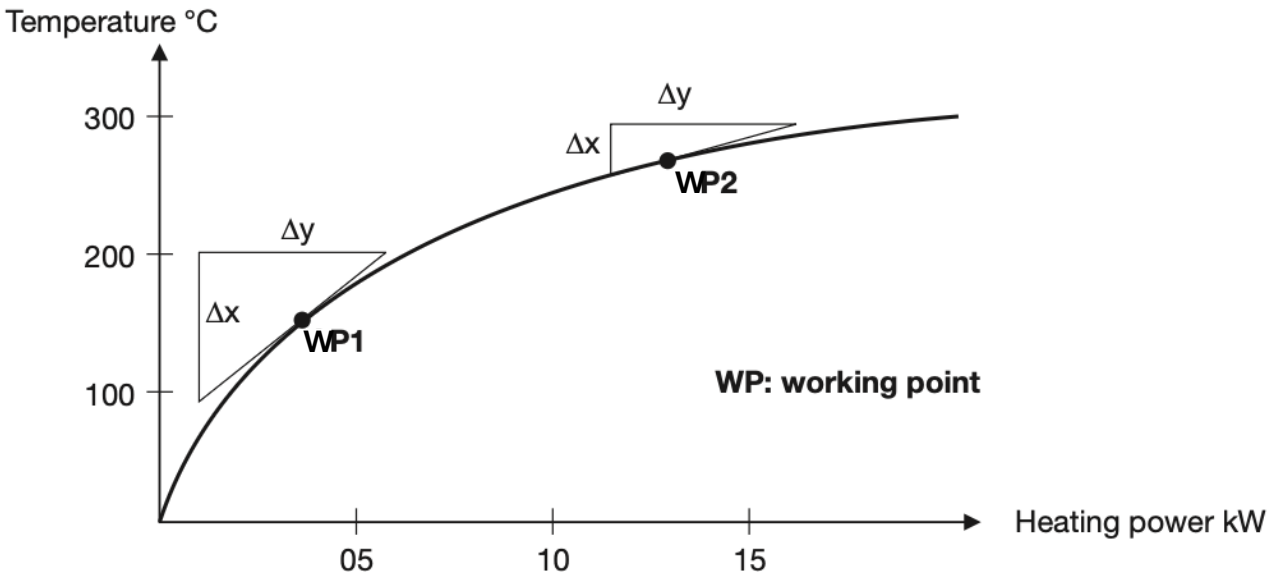


# Working Point and Linearization

- Controllers designed and tuned **around** the operating/working point of the system
- Linear approximation of nonlinear systems (often near equilibrium)
- Valid locally

Example: furnace




⇨ $K_S$ depends on the working point

$$\text{WP1: } K_S = \frac{\Delta x}{\Delta y} = \frac{100\ °C}{5\ kW} = 20\ \frac{°C}{kW}$$

$$\text{WP2: } K_S = \frac{\Delta x}{\Delta y} = \frac{30\ °C}{5\ kW} = 6\ \frac{°C}{kW}$$

# Proportional Band

Range of error over which the controller moves its output (the input of the process to be controlled) from 0% to 100%

Inverse of proportional gain: $X_p = \frac{1}{K_p} \cdot 100\%$

Smaller band → stronger control action (from 0% to 100% over a narrower error range)

Common industrial specification